Alonzo Cabrera
Portfolio: aloncs.github.io
Linkedin: https://bit.ly/463IGkP
alonzocabrera48@gmail.com
GitHub: https://github.com/AlonCS

Computer science student with hands-on experience in ROS 2 integration, robot teleoperation, and field data collection for navigation and manipulation models. Seeking robotics research internships focused on robot learning and vision-language models.

SKILLS

Languages: Python, JavaScript, C++, SQL, R, KAREL, HTML/CSS
Frameworks: PyTorch, React, ROS 2, MoveIt 2, OpenCV
Robotics & AI: CNNs (ResNet-18), Vision Language Models, PID Control
Tools: Git/GitHub, Docker, Google APIs, Excel, Autodesk Fusion, SolidWorks, CATIA, Isaac Sim, Gazebo
Platforms & Interfaces: Linux, Windows, Arduino, Raspberry Pi, NVIDIA Jetson, FANUC CRX25i, CAN bus, HMI Systems
Spoken Languages: English (native), Spanish (proficient)

WORK EXPERIENCE

Joby Aviation Marina, CA
Automation Intern February 2025 - May 2025

- Developed a KAREL diagnostic logger on a FANUC cobot replica, capturing position and velocity at set intervals to investigate faults interrupting robot operation.
- Automated thermocouple test analysis using Python and Excel, providing the automation team with an analysis interface and diagnostic reports for PID tuning.
- Shadowed a staff controls engineer during autoclave setup and drafted thermocouple, pressure-sensor, and control wiring documentation for a planned technician handoff in Watsonville.

Maniro AI Watsonville, CA
Robotics Intern May 2025 - August 2025

- Developed a ROS 2 teleoperation system enabling Xbox control of Amiga navigation and manipulation through a microcontroller and CAN bus, independent of farm-ng's software stack.
- Collected teleoperated demonstrations and evaluated navigation, strawberry-picking, and runner-cutting models on Amiga robots, supplying field data used to improve model accuracy.

Digital Nest Salinas, CA
Web Development Intern August 2024 - May 2025

- Redesigned websites for 5+ clients using Wix, Squarespace, and WordPress, adding functionality and improving readability and usability.
- Collaborated with designers and developers on project architecture, design implementation, and feature development using GitHub.
- Built the frontend for an internal scheduling application, visualizing randomized intern pairings and schedules generated from configurable rules.

Farm-ng Watsonville, CA
Sales and Education Intern July 2024 - December 2024

- Co-developed an Amiga robotics curriculum used across 40 high schools, translating platform training into instructional materials for educators and students.
- Engineered and tested a compact Amiga-based sprayer for operation between strawberry tables; the unit was sold to Driscoll's.

PROJECTS

Spacio|*ReactJS, Apps Script, Google Calendar API, JavaScript*
Back-end developer

- Built Google Calendar API integration to create, update, and cancel room reservations, with availability checks to flag scheduling conflicts.
- Integrated Google OAuth sign-in and session management, including session restoration and automatic logout when access tokens expire.

LEADERSHIP & ACHIEVEMENTS

- Mentor & Co-Instructor at X Academy, guiding students in MATE underwater robotics and autonomous farm robotics.
- Winner of the Western Growers specialty crops award, 2026 Farm Robotics Challenge (Division II).

EDUCATION

B.S. Computer Science — UC San Diego August 2026–Present
A.S. Mathematics and Computer Science — Hartnell College August 2024–May 2026